U.S. Flight Delay Propagation

Rahul Iyer, Michael Chian, Shaunak Warty, Sam Sukendro

Introduction

Current flight delay information is typically presented through airline reports, airport statistics, formulas, or basic dashboards showing averages such as delay times and cancellation rates. While useful, these methods **fail to capture** how delays spread across the broader flight network. A disruption at one airport can trigger delays elsewhere, causing passengers to miss connections, increasing airline operating costs, and contributing to passenger frustration nationwide. Existing research analyzes these effects mathematically, but offers limited interactive exploration. **Our project** addresses this gap through **interactive visualizations** that help users identify reliable flights, understand the structure of the U.S. flight network, and explore how delays propagate nationwide.

Algorithm Approach

We built a four-stage pipeline to clean, standardize, and reduce 26.3 million U.S. domestic flight records (2022–2025) to **25.9 million valid flights**. Aggregation was performed in **DuckDB**, generating summary tables for **377 airports, 7,565 routes**, hourly delays, airline performance, and monthly trends. These outputs were then used to construct the flight network and delay propagation model.

A key challenge was identifying when delays at one airport likely caused delays elsewhere using only public data. Unlike prior work that relied on private airline records or dense time-series data, our method uses a **novel temporal correlation algorithm** on open-source data.

We flag a propagation event when, on the same date and airline:

1. Flight A → Hub H departs **15+ minutes late**
2. A later flight H → B also departs **15+ minutes late**
3. Departures are **30–90 minutes apart**, which represents a realistic airplane turnaround window

Running this logic across the dataset found **9.66 million raw propagation events**, summarized into **6,695 hub-route relationships** with propagation frequency, average delay impact, turnaround time, and airline coverage.

Visualization Approach

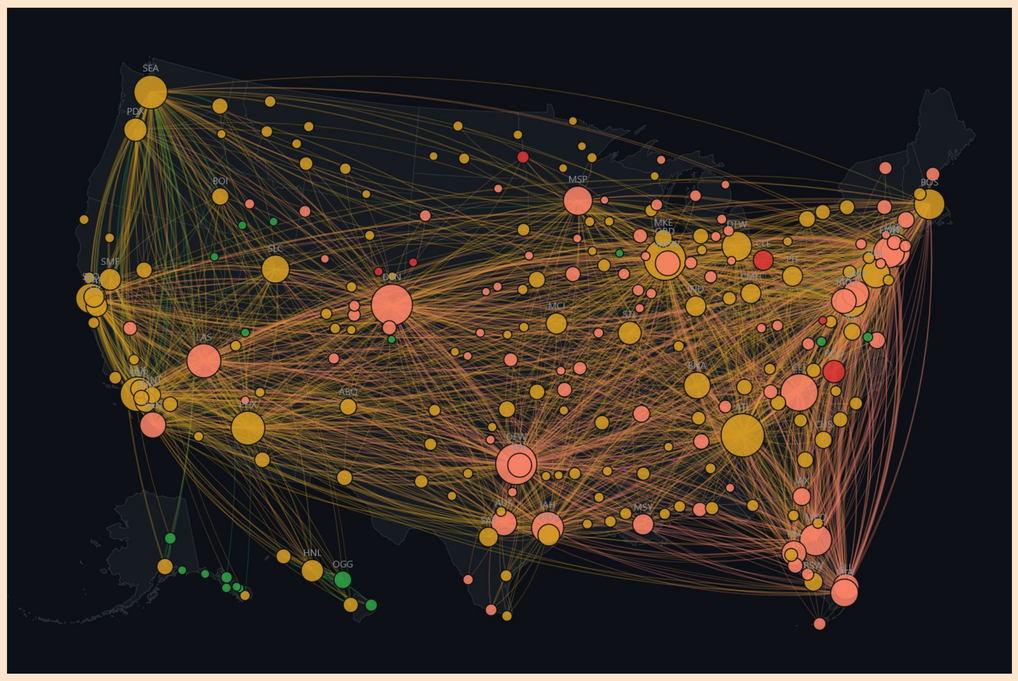
A FastAPI backend loads precomputed tables into memory, while the frontend provides four interactive views. Our system innovates by identifying network structure, hub importance, and cascading disruptions through interactive exploration that users can manipulate.

1. Network Map
Displays all 377 airports on a U.S. map. Node size represents flight volume, while color shows average departure delay. Users can filter by time of day to compare morning vs. evening congestion patterns.

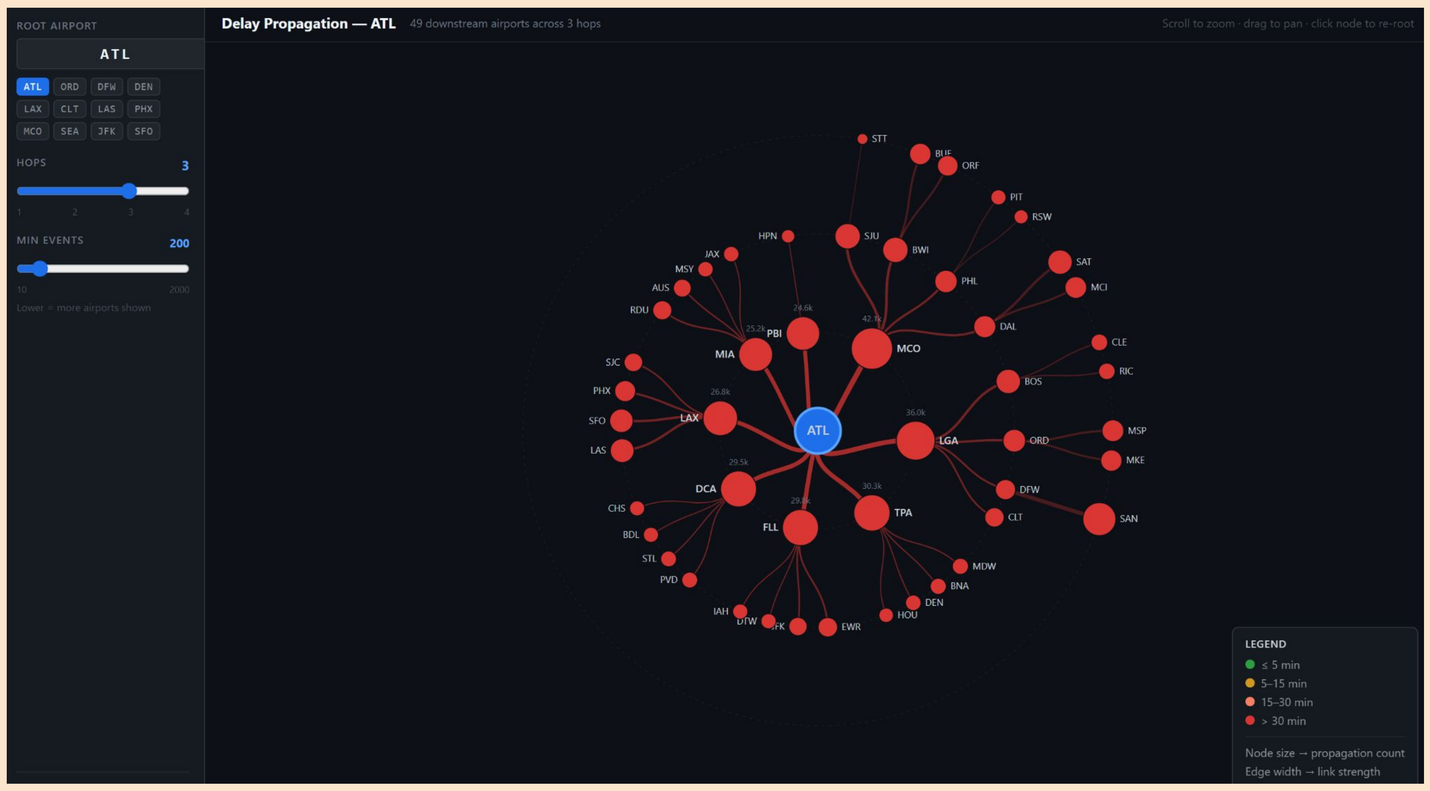
2. Airport Detail View
Provides a deeper look at any airport, including: network centrality metrics, hourly delay distributions, top airports it sends delays to, top airports it receives delays from.

3. Delay Propagation View (Novel Innovation)
An animated radial tree visualization shows how delays spread outward from a selected hub airport. The center node represents the selected airport while outer rings represent downstream airports by hop distance. Users can re-root the tree at any airport, filter weak links, and hover over airports to inspect specific metrics. This novel re-rooting interaction lets users follow delay chains dynamically, which static dashboards cannot do.

4. Timeline & Airline Comparison
Allows exploration of monthly and daily delay trends, plus reliability and propagation differences across airlines.



The Network Map view showing the **377 U.S. airports.** Darker red nodes represent airports with higher average departure delay times, while node size represents flight volume. DEN (Denver International Airport) stands out for being a major east-west hub with moderate delays.



The novel **Delay Propagation View** showing the **inferred delay propagations** from ATL (the root airport that the user has selected). From ATL, the next radial ring of airports represents the next set of airports where a delay at ATL will propagate to, and so on and so forth through concentric rings. Users can interactively add/remove outer rings, or adjust a threshold to filter weaker propagation links.

Data

Our project uses publicly available U.S. domestic flight performance data **downloaded** from the **Bureau of Transportation Statistics (BTS)**. The dataset contains detailed records for scheduled domestic flights from 2022 to 2025, including airports, airlines, departure times, arrival times, cancellations, and delay durations.

In total, the raw dataset contains **26.3 million flight records** taking up **1.61 GB** uploaded onto Google Drive, making it both large-scale and **temporal**. After cleaning invalid or incomplete entries, 25.9 million valid flights remained for analysis. The data spans 377 airports and 7,565 flight routes, allowing us to model the U.S. air transportation network and study how delays vary across locations, airlines, and time.

Experiments and Results

Our evaluation focuses on **three questions:** whether users can successfully navigate the system **without prior training**, whether the visualizations **clearly communicate** delay and propagation patterns, and whether the algorithm's outputs **align with real-world** aviation behavior.

User Study (Ongoing)
We are conducting a user study with peers in our graduate program. Participants use the application directly and complete four guided tasks: identifying the most influential propagation hub, tracing a delay cascade through the propagation tree, finding peak delay hours at Hartsfield–Jackson Atlanta International Airport using time filters, and comparing airlines by on-time performance. Afterward, participants complete a survey measuring task success, ease of navigation, interpretability, and overall usefulness. Final quantitative results are currently being collected and will be incorporated into the completed study.

Algorithm Validation
We evaluated the propagation model by comparing its rankings with known patterns in U.S. aviation. ATL and ORD (Chicago O'Hare International Airport) emerge as the strongest propagation hubs, consistent with their role as two of the busiest connecting airports in the country. EWR (Newark Liberty International Airport) also ranks highly, matching its well-known congestion and delay sensitivity. Peak propagation activity occurs in the late afternoon and evening, reflecting the real-world tendency for small morning delays to compound throughout the day.

Network Analysis Results
We also modeled the U.S. domestic system as a directed weighted graph with 377 airports and 7,565 routes. Network metrics produced realistic rankings:

1. ATL has very high degree centrality (0.86) and over 1.2 million incoming flights, confirming its central role in the network.
2. DEN (Denver International Airport) has strong betweenness centrality (0.164), reflecting its geographic importance as a bridge between eastern and western U.S. routes.
3. DFW (Dallas/Fort Worth International Airport) has a high PageRank (0.0412), indicating strong connections to other major hubs.

These results suggest our methods capture meaningful structure in both the flight network and delay propagation system.

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A key challenge was identifying when delays at one airport likely caused delays elsewhere using only public data. Our method uses a **novel temporal correlation algorithm** on open-source data that found **9.66 million raw propagation events**.

The intuition behind our approach is that aircraft routing creates a dependency chain (a late inbound plane physically cannot depart on time). Through our algorithm, we are able to approximate the conditions that allow for one delay to cause another.

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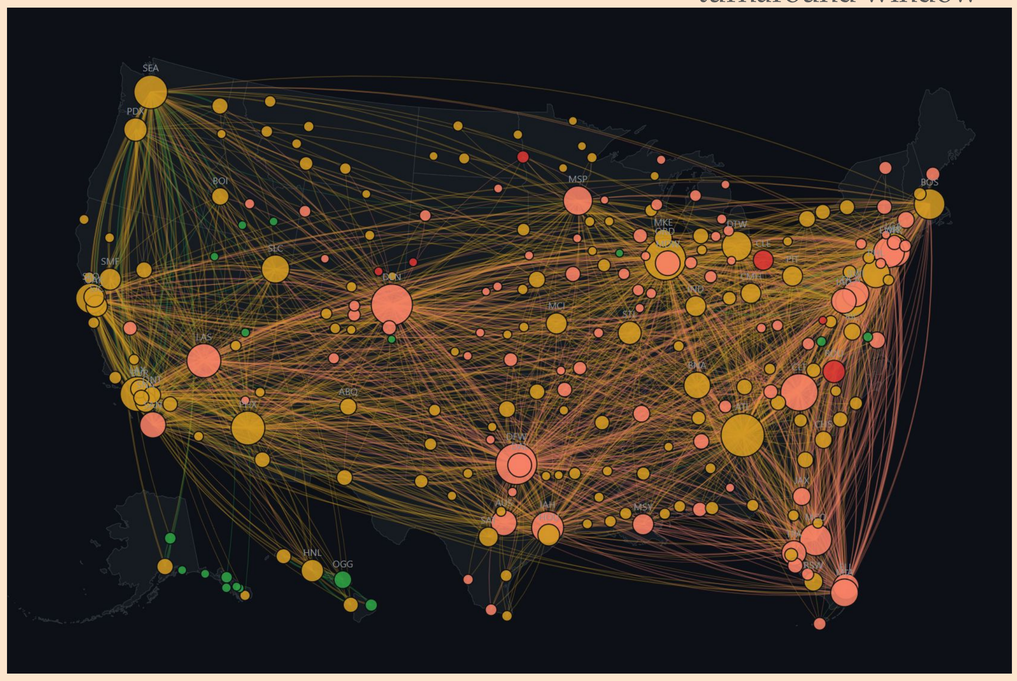
These views include the following: a **Network Map** mapping 377 U.S. airports, an **Airport Detail View** highlighting any airport with calculated statistics, a **Delay Propagation View (Novel Innovation)**, animating a radial tree visualization that shows how delays spread outward from a selected airport, and a **Timeline & Airline Comparison View** exploring delay trends across time and airlines.



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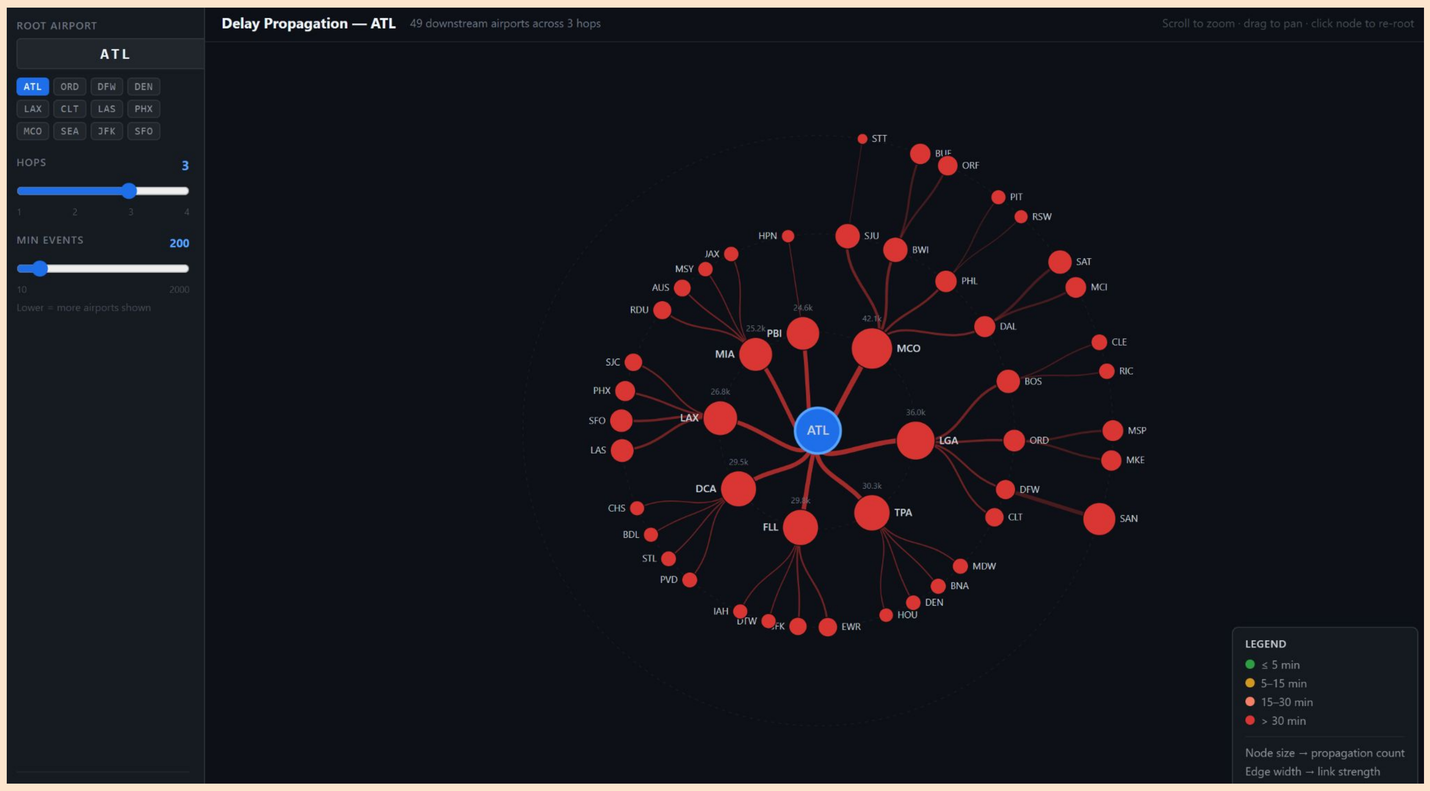

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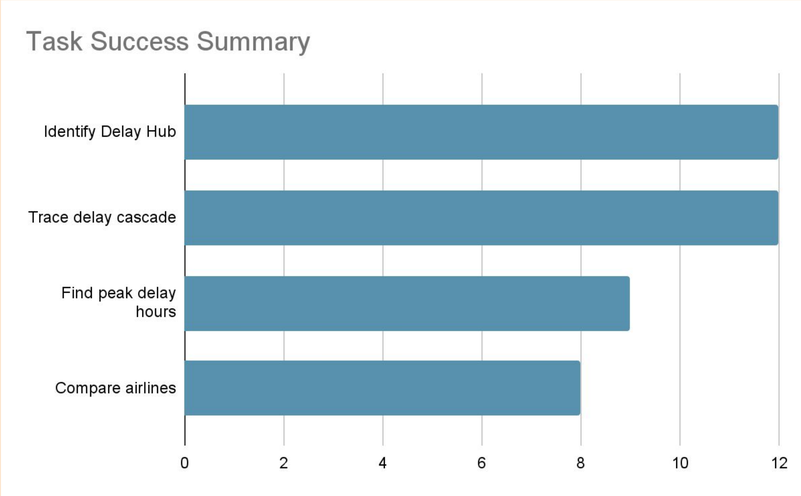
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